

School of Computer Science Engineering and Technology

Course- B.Tech.	Type- Core
Course Code- 2025CSET106	Course Name- Discrete Mathematical Structures
Year- I	Semester- Even (2 nd Semester)

Tutorial-3

Tutorial No.	Name	CO1	CO 2	CO3
3	Predicates and Quantifiers, translate English statements into logical expressions, prove logical equivalences between statements involving quantifiers.	✓	--	✓

Objective: The primary objective of this tutorial is to learn about Predicates and Quantifiers, translate English statements into logical expressions, and prove logical equivalences through statements involving quantifiers.

1. Use inference rules of propositional logic to check the validity of the following arguments.

- If I run, I will get there quicker. I got there quicker. Therefore, I must have ran.
- If I run, I will get there quicker. I did not run. Therefore, I did not get there quicker.
- If Bennett University is in Greater Noida, then Bennett University is in UP. Bennett University is in UP Therefore, Bennett University is in Greater Noida.
- Jatin is either a policeman or a footballer. If he is a policeman, then he has big feet. Jatin does not have big feet, so he is a footballer.
- If the company invests in renewable energy, it will reduce its carbon footprint. The company has decided to invest in renewable energy. Either the company reduces its carbon footprint, or it faces public backlash. Therefore, the company will not face public backlash.
- If a student attend lectures of DMS, student's understanding of computer science will improve. Students' understanding of computer science has not improved. If a student is not attending lectures, either student is not present on campus or DMS examination is not tough. Continuous evaluation policy confirms that the student is present on campus. Therefore, DMS examination is not tough.

2. Write the symbolic form of the following sentences without using the notion of universe of discourse (unrestricted domain).

- All lawyers are liars.
- Some lawyers are not liars.
- Any person who becomes a politician will get corrupt.
- No doctors know calculus.
- For every positive integer, there is a positive integer greater than it.
- Every integer is either positive or negative.

3. Write the symbolic form of the following sentences using appropriate Predicate and Quantifier.

- Every Indian festival is celebrated either based on solar or lunar calendars.
- Every citizen is eligible to vote if and only if they are above 18 years of age.
- In a family, every member either enjoys spicy food and dislikes sweets or likes sweets but prefer spicy flavors.
- Every even number greater than 2 can be expressed as the sum of two primes.

4. Translate each of the following predicate into an English statement.

- a) $\exists x: (s(x) \wedge \forall y: (s(y) \rightarrow (x = y)))$, where $s(x)$: x is a book on the table.
- b) $\forall x: \text{prime}(x) \wedge \text{greater}(x, 2) \rightarrow \text{odd}(x)$.
- c) $\exists x \forall y: x + y = y$, provided the universe of discourse consists of all real numbers.

5. Consider the following predicates:

P (x, y): $x > y$

Q (x, y): $x \leq y$

R(x): $x - 7 = 2$

S(x): $x > 9$

If the domain of discourse is the real numbers, give the truth value of each of the following propositions:

- a) $\exists x: R(x)$
- b) $\forall y: \neg S(y)$
- c) $\forall x \exists y: P(x, y)$
- d) $\exists y \forall x: Q(x, y)$
- e) $\forall x \forall y: (P(x, y) \vee Q(x, y))$
- f) $\exists x: S(x) \wedge \neg (\forall x: R(x))$
- g) $\exists y \forall x: S(y) \wedge Q(x, y)$
- h) $\forall x \forall y: (R(x) \wedge S(y)) \rightarrow Q(x, y)$

6. Use predicate logic to check whether the following arguments are valid or not.

- a) All foods that are healthy to eat do not taste good. Tofu is healthy to eat. You only eat what tastes good. Therefore, you do not eat tofu.
- b) Everyone from Bennett University lives within 50 km of the India Gate. Someone from Bennett University has never seen the India Gate. Therefore, who lives within 50 km of the India Gate has never seen the India Gate.
- c) All protons are subatomic particles. All neutrons are subatomic particles. Hence, all protons are neutrons.

7. Write the symbolic form of the following sentences using appropriate Predicate and Quantifier, and then find the negation for the same.

- a) Every student is honest.
- b) There is a teacher who teaches Mathematics.
- c) Every employee reports to some manager.
- d) There is a person who likes everyone.
- e) Everyone has a friend.